

11. Explain the goals and objectives of Environmental Education (B) [Ans: 1:11 + 1:12]
12. Explain the need for Environmental Education in the present Indian Context. (B) [Ans: 1:13]
13. Briefly state the importance of Environmental Education. (B) [Ans: 1:14]
14. What is Environmental Education? Explain its scope. (A) [Ans: 1:08 + 1:15]

Unit - II

NATURAL RESOURCES, PROBLEMS AND SOLUTIONS

2:00 Introduction

For man to live air, water, land, sunlight and food are the most indispensable ones. Man gets his food from plants and animals. Agriculture, fishing and cattle-rearing constituted the occupations of the early man. As civilization started progressing, man started using minerals and energy resources to expand the comforts of his life. In the present day life style, we are pushed to that stage to think that without electricity human life is not possible even for a minute on this planet called earth. Due to the continued use of natural resources, different problems including environmental pollution have shot up. In this lesson, we are going to discuss the various kinds of natural resources like water, land, air, forests, food and energy and the problems associated with them.

2:01 Various Kinds of Natural Resources

The following are the various kinds of resource available in nature :

- i) Land Resources
- ii) Forest Resources

- iii) Water Resources
- iv) Mineral Resources
- v) Food Resources
- vi) Energy Resources

We shall discuss each of these natural resources and problems associated with them.

2:02 Land Resources

Land is one of the most important natural resources in India. Out of 510 million sq. km. of the world total area, land accounts for nearly 29% (149 million sq. km.) and is distributed in the form of continents like Asia, Europe, Africa, North America, South America and Australia. In this, India has 2.4% of the land area of the world. Based on height variations, land surface area of the earth is classified into a number of land forms like mountains, hills, plateaus, plains and valleys. Out of the total area of the world, 25% is in the form of mountains, 45% as plains and 30% as plateaus. Of these land forms, riverine plains and coastal plains play a vital role in human life. Nearly 66% of the world population is concentrated in the plain regions, mainly because of favourable living conditions.

Of the total world land area, 1% exists as human habitats, 11% as agricultural lands, 24% as grazing lands, 33% as forests, 28% as rocky hills, deserts

and swamps and 38% as reserved areas for parks and wild life sanctuaries.

Soil is very important environmental component for human, animals, and plants. It is the basic medium for food, vegetation and natural resources. It is also the base for the development of human culture and civilization. There are various aspects and components of soils - structure, texture, moisture, profiles and horizons.

Land is highly useful to obtain the basic necessities like food, clothing, and shelter. By utilizing land, man is able to do the most important occupations like agriculture, industries, transportation, mining, cattle and sheep rearing. Hence, the role of land in the day to day affairs of man is said to be vital.

2:02:1 Factors Affecting Land Resources

Among the factors affecting land resources the important ones are : (i) Soil erosion (ii) Landslides and (iii) Desertification.

2:02:1:01 Soil Erosion

A) Meaning of Soil Erosion

Soil erosion is the drifting of the top layer of the soil from one place to another. It is a natural process. Loss of vegetation (Plants and Trees) makes land more susceptible to erosion.

B) Reasons for Soil Erosion

Generally wind and water are responsible for soil erosion. Apart from these natural factors, there are some man-induced factors responsible for soil erosion. Important among them are : (i) Destruction of forests (ii) Mining operations (iii) All developmental activities like construction of buildings, dams, bridges, industries, recreational parks etc. which ultimately ends up with using substantial area of land, leading to soil erosion (iv) over-grazing of cattle leaves the land barren and makes it susceptible for soil erosion by wind and water.

C) Effects of Soil Erosion

- i) Floods
- ii) Aridation (drought),
- iii) Depletion of nutrients in soil rendering it infertile

There are other effects also which are detailed here under :

- iv) Thinning of soil layer reduces the plant growth, nutrients and water
- v) Reduction in crop production
- vi) Disturbance in air-humidity balance
- vii) Disastrous effect on micro-organism
- viii) Loss of underground water

- ix) Effects on seasonal and perennial water bodies.
- x) Addition of run off soil and mud leads to reduction in absorption of water in lakes and ponds.
- xi) Fish cultivation becomes difficult due to erosion of soil.
- xii) Alkalinity and seepage increase in the plains.
- xiii) Hydraulic machines and other pumping machines get rusted due to this mud or soil water.

2:02:1:02 Landslides

A landslide is the descent of a mass of earth and rock down a mountain slope. Like earthquake, flood, volcanic eruption, land slide too is a natural disaster. Generally landslides occur due to continuous heavy rain. Soil erosion, inadequate draining of rain water and excessive pressure on waterbeds cause landslides. Landslides occur mostly in rainy season.

Activities of man too contribute to the cause of landslides. Constructing buildings on hill or mountain slopes preventing the free flow of rain water and laying a number of roads by levelling mountain rocks lead to landslide. Landslides that suddenly rush down a steep slope can cause great destruction across a

wide area of habitable land and sometimes cause floods by damming up bodies of water. Landslides cause obstructions to road and rail traffic; sometimes the whole village gets buried due to a landslide.

By strengthening the hill slopes, landslides could be avoided. At the bases of hill slopes concrete check walls could be erected. By constructing draining canals to long distances, rain and rivelet waters could be safely carried away from the hill slopes. By adopting these effective measures to prevent landslide, damages to men and materials could be avoided to a large extent.

2:02:1:03 Desertification of Land

Deserts are unused land areas of the world. When a region remains without any rainfall permanently it is referred to as a desert. In deserts, as there is no humidity, hot winds blow through the day raising the temperature of the atmosphere. [E.g. Western Rajasthan and Thar desert in India, Sahara desert in North Africa, Kalahari in South Africa, Colorado in North America, Atacama in South Africa, West Arabian desert etc.]. Deserts are mostly sandy; no vegetation is possible; density of population is also very low.

Man has become the most important factor for the extension of desert areas. Nature is also responsible for desertification. Selfish attitude of man

and his activities like deforestation, overgrazing, excessive utilization of ground water, pollution of water bodies, over population and mining operations in a large number of places, lead to the land becoming desert in course of time. Man is also responsible for climate changes which lead to the formation of deserts.

Desertification starts from a small fertile land area. Large amount of dust particles are added to the atmosphere, leading to drought conditions. If the drought conditions prevail for a longer period of time in a moist region, it may lead to the formation of desert in course of time. Deforestation is mostly responsible for increasing temperature and scanty rainfall, producing atmospheric drought conditions in any place which may lead to the formation of desert i.e. temporary drought conditions become permanent.

2:02:2 Prevention of Desertification

Following measures are suggested to prevent the process of desertification.

- i) In border areas of deserts, sand dunes should be prevented from advancing by growing thick vegetation. Mostly drought resistant plants like xerophytes be planted in the border zones.
- ii) Plants may be grown in desert regions by taking water from the rivers through canals and resorting to drip irrigation as it is done in Israel.

- iii) The use of underground water near the desert region should be properly regulated, so that moisture in the sand will not be reduced to a greater extent. Soil erosion by wind over the sandy areas may be prevented.
- iv) By controlling air pollution, rise in atmospheric temperature may be prevented thereby increasing the chances of getting more rainfall.
- v) Reducing the use of ground water, desalination of sea water could be undertaken to provide drinking water.
- vi) Awareness should be developed among the people living in desert regions the necessity of preventing the process of desertification.

2:02:3 Meaning of Soil Fertility

Soil is the uppermost part of the earth's surface. It is being modified by physical, chemical and biological process over time. Soil is formed over long periods of time. It takes 600 years to form 1 inch of the top soil. It is an essential medium for plant growth. It is a dynamic mixture of solid and dissolved mineral matter, living and dead organic matter, water and air.

Soil fertility is the ability of the soil to supply essential nutrients to plants. Generally, soils that have a high proportion of clay and organic matter have higher fertility. The water-holding capacity of a soil is determined by its texture. Decrease in soil fertility is referred as 'Degradation of land resources'.

2:02:3:01 Causes for the Degradation of Land Resources

1. Soil Erosion : Due to soil erosion the upper layer of earth is removed resulting in depletion of nutrients in the soil.

2. Flood and Water-logging : Flood causes soil erosion and thereby depletion of nutrients in the soil. Water-logging causes the flushing out of minerals from the soil.

3. Use of Chemical Fertilizers and Pesticides : Continued use of chemical fertilizers destroys the micro organisms in the earth which leads to decrease in soil texture. Further, it increases water-logging in fields, leading to salinization.

4. Cultivation Without Crop Rotation : In cultivating a single variety of crop, year after year the crop uses up particular type of minerals from the soil and produce high yields initially for some years. Afterwards soil fertility decreases due to exhaustion of that particular type of minerals in the soil, leading to land becoming barren. To avoid this, crop rotation method should be adopted. E.g. after Paddy cultivation, we can raise corns followed by growing vegetable plants for a short period; then legumes may be grown and mulching may be adopted to stabilise nitrogen in the land.

2:02:3:02 Conservation of Soil

Conservation means prevention of wasting natural resources. Soil conservation means the prevention of wastage of soil and depletion of earth. Due to proper conservation methods not being adopted, vast areas of fertile land have been converted into waste lands. Some conservational methods are as under :

(1) Reforestation : The hilly areas in which the soil erosion is maximum need soil conservation measures. Reforestation means planting of trees which are quick-growing and provide efficient plant cover for prevention of soil erosion. Planting of shelter belts and other wind breaks prevent soil erosion by winds. The plant species like *Saccharum munjo*, *Cynodon dactylon* etc., are recommended for desert areas to check sand shifting.

2) Agronomic methods : These include (a) contour planting or ploughing at right angles to the slopes to check flow of water down the hills avoiding soil erosion.

b) *Contour furrowing* : Furrows should be made on slopes to hold the run off water and soil.

c) *Strip cropping* : These are alternate strips of densely growing crop covers which prevent soil erosion.

d) *Mulching* : Plants and animals are not burnt but after decomposition used as humus.

e) *Lay farming* : Crop-grass rotation in agriculture is called lay farming which checks soil erosion and holds soil particles in the roots of grasses.

3) Terracing : Terracing means the division of sloppy areas into series of small flat fields by means of ridges when placed in such a manner that they hold water. These absorb water and thus check run off. This breaking up of long slopes into small water sheds channelises water to the sides of fields thus preventing soil erosion.

4) Construction of Dams : Floods are a regular feature every year in rivers. Floods are the major cause of soil pollution and erosion. This heavy erosion can be checked by constructing dams and these dams provide water for irrigation and in the process infertile land can be converted into fertile ones.

5) Controlled grazing : Excessive grazing renders land barren and the soil particles thus exposed become dry. The underground roots become weak due to lack of foliage and the soil becomes dry and weak which is easily eroded by wind and water.

6) Soil fertility : The fertility of the soil should

be maintained by green manuring and using bio-fertilizers to restore the nutrients. Only fertile soil supports crop which ultimately adds to the inorganic and organic constituents of the soil. **Indian Agriculture Research Institute, New Delhi**, listed three steps (i) that the country should be divided into soil regions, ii) detailed surveys be conducted in 25% area (iii) soil depth, texture should be surveyed. Maintenance of soil conservation measures are more important. We cannot afford to ignore these steps. This is one of the major problems of Indian agriculture. More than about 100 million hectares of land is still exposed to this hazard. About 40 million hectares of such land is under agriculture. Water erosion is common in Gujarat, Maharashtra, Karnataka, Andhra Pradesh, Punjab, UP, Bihar. Wind erosion is common in Rajasthan, Punjab, MP, UP etc.

National Land Resource Conservation and Development commission remarked that uncontrolled grazing, felling of trees, shifting cultivation and faulty agricultural methods are the main causes for the loss of fertility of land.

2:03 Forest Resources

The term 'forest' has been derived from the Latin word '**Foris**' which means 'outside'. Forests are self sustaining wooden tracts with biotic community dominated by trees and plants. They contain 90% of

the terrestrial biomass and cover 40% of the land. Forests occupy 63.85 million hectares (19.48%) out of 329 million hectares of geographical area of India. In India there are four major types and sixteen sub-type forests. Our forests meet 40% of the energy needs of the people and 30% of the fodder needs of the cattle population of our country.

It is said, only those nations which contain forests in one third of their geographical area, are fertile and receive copious rainfall. Both the departments of Indian Government viz **Indian Forest Survey** and '**Indian Remote Sensing**' insist that the forest area is to be increased from the present 20% to 33%. Former Central Minister **Mr. Mohan Dhar** founded an organization named '**Vanroy**' in Poona and silently worked for afforestation. According to him, shortage of rainfall in our country could be avoided if we raise forests in 110 million hectares.

2:03:1 Uses of Forests

Direct Uses

- i) Forests provide domestic fire wood, timber to make furniture and building materials like doors and windows. Forest products can also be used to produce sandal oil, paper boards, newsprint, cloth, match stick, agarbathi stick etc.
- ii) They also fulfill the human food needs to some

extent, providing fruit trees like almond, walnut, cashewnut, chilgoza as well as honey.

- iii) They provide some herbal spices like nutmeg, cinnamon and clove which help us curing human ailments.
- iv) Certain chemicals used in dyeing industries are obtained from the barks of some trees. They also provide gum, natural rubber etc.
- v) Some kind of mineral deposits and valuable stones like diamond are obtained from forests.

Indirect Uses

- i) Forests protect the soil from the direct action of rain and wind, as plants and trees hold the soil particles firmly. Thus soil erosion is prevented.
- ii) Forests have a moderating effect on temperature and other climatic conditions. They increase the rainfall frequencies and the level of humidity in the atmosphere.
- iii) Forests reduce atmospheric pollution by absorbing polluting gases particularly CO_2 and collecting suspended particulate matter.
- iv) Forests offer food and shelter for many a wild animal and plant. Plant litter and humus make the soil spongy, helping to hold rain water and prevent run off. Rain water slowly percolates into the soil providing perennial supply to springs and rivulets.

2:03:2 Deforestation

Deforestation is deliberate removal of forest cover by felling trees and burning plants. In India, one third of land was covered with forests during the late 1930s; but now it got reduced to 19.48% of land. There are many factors responsible for deforestation. Among them, conversion of forestland for (i) creating human habitats (ii) undertaking developmental projects like construction of dams and hydroelectric power plants, laying roads and rail tracks (iii) shifting cultivation (iv) industrial demand of wood (v) mining operations (vi) demand for domestic fuel and timber for furnitures and fittings and (vii) over-grazing are the major ones. Forest fires are also responsible for destruction of forests.

To protect forests, Indian Government legislated "Forest (Conservation) Act in 1980 and amended it in 1988 to ensure heavy punishment for the offenders.

2:03:3 Reasons for Deforestation

- Rapid population growth that increases the pressure on the limited resources available.
[e.g. Demand for land for (i) cultivation (ii) commercial agriculture like planting tree crops such as rubber, oil palms, Sandalwood, coffee

etc. (iii) construction of roads, rail tracks, dams and hydro-electric power plants (iv) Live stock grazing (v) meeting the demand of firewood fuel and vi) meeting the demand of timber for household uses and for industries like paper and textile mills, match stick and Agarbathi manufacturing units]

- Frequent forest fires, either natural or man-induced.
- Flawed government policies for managing forests
- Inadequate understanding of the costs and benefits of deforestation resulting in faulty economic analysis.
- Colonial patterns of resource exploitation that emphasize on maximum short term gains.
- Debt-load carried by many developing countries that is forcing them to resort to timber exports (Sweden supplies 30% of Europe's timber and 60% of paper pulp)
- High demand in developed nations for tropical timber and other commodities.
- **Shifting Cultivation** : A patch of land is cleaned through looping and burning. The ash is mixed with soil. The cleared land is cultivated for 2 to 3 years. During this period the soil loses all its nutrients through run off. Absence of organic matter reduces moisture retention and increases

soil erosion. Permanent degradation occurs if shifting cultivation is carried on. In short intervals, weeds may grow in such areas. It is estimated that more than 5000 Sq. Km. of forests are cleared annually through shifting cultivation.

2:03:4 Adverse Effects of Deforestation

- i) Forests absorb rain water like a sponge and slowly release it when needed by the soil. But in the absence of forests, rain water is wasted as "run off", and it flows into the sea or it is evaporated from land.
- ii) Deforestation also leads to the erosion of the rich top soil, frequent floods, droughts and desertification with decline in agricultural productivity and non-timber products.
- iii) Alterations in the global climate result due to the changes in the earth's reflection of light and heat from the earth's surface, when the light-absorbing forests are removed. The heat balance of the earth may produce changes in wind and rainfall patterns.
- iv) Deforestation results in the building up of carbon dioxide in the atmosphere, which ultimately leads to global warming. This is called '**green house effect**'.

- v) Deforestation results in the loss of an important sink for ozone.
- vi) Removal of forests denies the storage of carbon in plant tissues or the absorption and breaking of pollutants.
- vii) Deforestation leads to the loss of habitat of various wild animals and destruction of plants and trees (**Extinction of flora and fauna**).
- viii) Deforestation results in the non-availability of timber and fuelwood for domestic and industrial use
- ix) Most of the tribal communities live in forest areas. Deforestation leads to the destruction of tribal life.

2:03:5 Measures to Prevent Deforestation

1. Political interference in policy matters like denotification of various parts of sanctuaries, providing mining and quarrying rights, permission for laying roads etc., should be done away with.
2. There should be adequate and less time consuming legal procedures that do not allow the miscreants to go scot-free.
3. There should be proper coordination among the

- various government departments to avoid delays in the settlement of various issues like disputes over the rights of the land and title.
- 4. The nodal officers and departmental officers should be adequately motivated to check deforestation.
- 5. New proposals for the establishment of hydroelectric, irrigation, mining, road and rail, thermal and transmission projects etc., should obtain clearance from Environmental Management Authority.

Multi pronged pressures on forests come from the growing population, cattle grazing, fuel and fodder collection, industry etc., Because of these pressures on the forest, the actual extent of forest cover in our country has dwindled to just 11% against the desirable 33% of the total land. Until the late seventies of the last century, forest land was a prime target for resettlement, agriculture and industrialization. The Joint Forest Management (JFM) introduced in many States in the early 1990s has resulted in the improvement of forest cover in the country to reach the present level of 19.48%.

2:03:6 Afforestation

Growing trees as a crop to develop forests in those places where previously no forest

existed is called afforestation, whereas reforestation refers to develop the cleaned forests into a full fledged forest cover. To achieve this, Government of India brought forward "**The Indian Forest Policy, 1988**". Under this policy the following rules and regulations were framed and implemented.

- i) Raising new forest cover in watersheds, slopes and other ecologically fragile areas.
- ii) Growing industry oriented trees like Teak, Casurina, Oak etc. which do not require much water, on fallow and free grazing lands.
- iii) Implementing 'Social Forestry' scheme in which through **Forest Development Institute**, fast growing tree saplings are developed and distributed to the general public.
- iv) Encouraging urban forestry.
- v) Encouraging Agro-forestry.
- vi) Helping the tribals to collect fuel, timber, fruits, honey, perfumes etc. without injuring the forest.
- vii) Advancing silviculture.
- viii) Grazing and litter removal.

2:04 Water Resources

Water is most important natural resource. It is vital for the maintenance of all forms of life and vegetations. We depend on water for irrigation, industry, domestic needs, drinking purpose, for sanitation and disposal of waste. Our water bodies are ponds, lakes, sea, rivers, oceans which have become polluted due to industrial development and urbanization.

70% of the Earth is covered by water and that is why it is called the 'Blue Planet'. Only 2.5% of the world's water is fresh while 97.5% is found in our oceans. Only 0.3% of fresh water is available from rivers, lakes and reservoirs; 30% from ground water; the rest is stored in distant glaciers, icesheets and inaccessible areas.

In India, the available fresh water from both surface and ground water is 5 million hectare metres and taking the average of 1500 cubic metres per year per person as the water need, the population of India will fall in the category of 'Water Stress'. At present ground water caters to about 5% of the irrigation requirements and 80% of domestic and sizeable portion of the industrial requirements.

2:04:1 Water Resources in India

India has the following water resources —
(i) Rain Water (ii) Surface - Flow Water and (iii) Ground Water.

2:04:1:01 Rain Water

The 'Hydrological Cycle' begins with evaporation from oceans in which water changes from liquid to vapour and enters the atmosphere, which due to chemical changes get condensed to form precipitation that falls to earth as rain, snow or hail. Rain water falling on earth, first percolate into the soil and then down to the water beds under the shallow rocky layers. The run off rain water reaches the ponds and lakes on earth surface or joins the canals or riverlets to transform into a river and finally reach the sea. India receives 70% of its rain fall through South West monsoon lasting from June to September and the rest 25% from Northeast monsoon during October to December. Amount of monsoon rainfall received differs from year to year, as well region to region. In our country, **Chirapunji** in **Megalaya** receives the highest rainfall (11,000 mm per year) and the lowest rain fall occurs in Western Rajasthan (100 mm per year). Whenever monsoons fail, water scarcity occurs in India. Deforestation is

said to be the major reason for decrease in rainfall and drastic climate changes. Rain water could be conserved effectively by erecting rain water harvesting structures in every building and the run off water could be deposited by constructing dams and check dams across the rivers.

2:04:1:02 Surface-Flow Water

Lakes, Ponds and Rivers are referred to as Surface-flow Water resources, Rainwater is the source for all the lakes, ponds and rivers. Ganges, Yamuna, Sindhu, Brahmaputra, Godavari, Krishna, Narmada, Tapi, Mahanadi, Vaigai and Cauvery are the major rivers of India. Among these the rivers Ganga, Yamuna, Sindhu and Brahmaputra originate from the higher altitudes of the Himalayas and the water flow in them is due to melting of snow and icebergs. These rivers are perennial rivers.

2:04:1:03 Ground Water

Water deposits found at a depth of about 1000 metres, from the earth surface are known as ground water. This could be pumped out through bore wells and used for domestic and industrial water supply as well as for irrigation. Ground water is the source of life in many regions of the world. It is very pure and potable until not polluted by man. Even

though the water table level diminishes during the dry seasons, it fulfills the needs of people.

Of the rain water received on earth, a small portion is lost by evaporation, a significant portion runs off and reach ponds, lakes and rivers; the remaining portion slowly percolates to the subsoil beds. Ground water is the part of sub-surface water. The hard upper surface of earth consisting of top soil layers and rocky subsoil layers, contains two zones — (i) Aeration Zone and (ii) Saturation Zone. Water percolated and spread in these zones forms the ground water deposits.

2:04:2 Uses of Water and its Over-exploitation

Water is an essential need for human life. 70% of the available water is used for agriculture, 24% and 6% are for industrial and domestic uses respectively. As the world population increases, water need also increases. But monsoon-failures have become too common nowadays. While developing countries are taking all efforts to expand and conserve their water-sources, in India we are allowing our water resources to dry up without any concern or care. Surface water sources get increasingly polluted day by day. We have started overexploiting

ground water through bore wells to meet the water demand for agriculture, industrial and domestic use. If this situation persists for long, then desertification will spread and people quarrel among themselves for a pot of water. So there is an urgent need for the Government and people to find a solution for water scarcity.

2:04:3 Ways and Means of Controlling Water Scarcity in India

India has three kinds of water resources – (i) Rain Water (ii) Surface-flow water and (iii) Ground Water.

Water scarcity could be avoided to a great extent by conserving the above referred three kinds of water resources and ensuring economic consumption of water in our daily life. For most of the Indian rivers, rain water constitutes the primary source. Rivers like Ganga, Yamuna, Sindhu and Brahmaputra as they originate from the higher altitudes of the Himalayas, they remain perennial. For ground water also, rain constitutes the primary source. Therefore protecting and conserving rain water obtained during the South West and North East monsoons is the most important means of controlling water scarcity in India. By interlinking the major rivers in our country, we can

easily tackle the twin problems of drought and flood. But as on date, the project of '**Interlinking Major Rivers of India**' remains on paper and no concrete step has been initiated.

Copious rain water received during the monsoon seasons could be conserved and stored by constructing dams across the rivers, constructing numerous checkdams here and there across the waterways, constructing recharge wells along the roadsides, roof-top rain water harvesting arrangements in urban dwellings and multistoried apartments and diversion of run-off water into existing surface water bodies like ponds and lakes are the measures suggested to tackle water scarcity. Disilting of existing water storage systems like ponds, wells, lakes and dams go a long way in the conservation of water as they increase the capacity of storage of water resources.

Preventing pollution of water and its sources, recycling of waste water from the household kitchen and bathrooms and use it for vehicle - washing and watering plants in the garden are the other measures suggested to conserve water by economising human consumption of water.

2:05 Mineral Resources

Mineral resources could be broadly be defined as 'Substances, Chemical compounds, rock concentrates having a definite physical and chemical composition in a form that can be refined or extracted to be used as a commodity'.

Mankind depends on mineral resources for various activities of day to day life. Glass tumblers, tools and equipment, utensils we use are all made of minerals. Predominantly minerals are inorganic in origin (e.g. Metals like Gold, Platinum and Iron, non-metals like Gypsum, diamond, mineral salts); but a few minerals like petroleum, amber and coal are organic in nature. Minerals are extracted from the earth in two ways. (i) Open pit type mining (ii) Closed tunnel type mining. Among the two, the latter type is extensively used all over the world. Mineral refining industries are generally located nearby the respective mines. While extracting minerals, we get 10% tar from them which finds use in many ways. Mineral resources belong to the non-renewable category of natural resources. Generally minerals are found in the form of rocky layers under many kilometers of depth from earth's surface. They get formed by geographical processes over a period of hundreds of years.

2:05:1 Categories of Mineral Resources

Minerals may be classified into the following three categories :

- i) Metallic Minerals
- ii) Non-mettalic Minerals
- iii) Fuel Minerals

2:05:1:01 Mettalic Minerals

Metallic minerals like Gold, Chromium, Nickel, Platinum etc. are found as particles embedded in hard rocks. These rocky layers are cut and brought above the surface and mettalic mineral particles are extracted from them. Metals like Aluminium, Manganese, Iron, Copper etc. are not found as mettalic particles but found in the form of their respective compounds. From them, the respective metals are isolated and extracted by employing different methods like smelting, roasting etc. Iron, Aluminium, Chromium, Manganese, Titanium and Magnesium are abundant and while other mettalic minerals like copper, Lead, Zinc, Tin, Gold, Silver, Platinum, Uranium, Mercury, Molybdenum are scarce.

2:05:1:02 Non-mettalic Minerals

Non-mettalic minerals range from silicate

minerals to sand, gravel, salts, soils, diamond, limestone, asbestos, Gypsum etc. These kinds of mineral resources are found in Karnataka, Rajasthan and Bihar. Gypsum and Mica are found in Tamilnadu. Diamond is found to some extent in Madhya Pradesh.

2:05:1:03 Fuel Minerals

Coal, Petroleum and amber belong to the category of fuel minerals. It is more appropriate to call them organic resources. In India Coal is available in abundance, particularly in West Bengal, Orissa, Bihar, Assam, Madhya Pradesh, Maharashtra, Meghalaya, Nagaland and Uttar Pradesh. Lignite (Brown Coal) is available in Neyveli, Tamilnadu.

Petroleum is found in Assam and Gujarat States as well as in Godavari basin and Bombay high seas. Plutonium is found in the Trivandrum beach.

Mineral resources are finite and at certain point, will be exhausted. In that case, there are four possible solutions.

- i) Explore for more minerals.
- ii) Recycle what has already been extracted and is in use.

- iii) Find out an alternative or substitute for a mineral in demand.
- iv) Design technologies which avoid that mineral.

2:05:2 Uses of Mineral Resources

- i) Gold, Silver and Diamond are used in the manufacture of jewels. Copper, aluminium, iron and nickel are used in the manufacture of various household articles, utensils, fittings and fixtures in industry, transmission of electricity etc.
- ii) Diamond, Platinum and many other metals find many industrial uses, in the production of various materials.
- iii) Aluminium is used to produce the body of aeroplanes and automobile engines.
- iv) Coal and petroleum are used as various kinds of fuel from cooking to aviation.
- v) Without the use of iron no weapons and tools, room fixtures and fittings, vehicles etc. could be manufactured. That is why **iron is referred to as black gold**, indicating its high value.
- vi) Various inorganic minerals find extensive applications in the production of chemicals, fertilizers and other industrial products.

2:05:3 Adverse Effects of Mining on Environment

Minerals are the rich natural resources of any country. Many industries have undertaken the task of extraction of minerals and refining them. They use large quantities of water and release enormous wastes, polluting the environment. Mining inflicts heavy damage to the environment and degrades its quality in the following ways.

(i) Air Pollution

Blasting the rocks, using drilling machines, transport vehicles employed in and around the quarries etc. contribute for the increase of dust and suspended particles in air. When rocks are blasted using bombs, emission of smoke in air becomes inevitable. Similarly drilling operations in coal beds emits poisonous methane gas. Thus in and around the mines, atmospheric pollution is very high.

(ii) Water Pollution

Water discharged from the mines, percolates into the earth, polluting the fresh water aquifers due to which people are affected by various diseases. Agriculture and live stock too are affected by mining.

(iii) Land Pollution

It is the land which suffers the most due to mining. During mining operations, upper surface of the earth is removed and heaped as piles. The area around a mine becomes denuded of vegetation, laced with stagnant pools of water or covered with mine tailings or slag. To produce few tones of gold, million tones of earth are to be removed. This kind of operations cause soil erosion and earth tremors. Once the mining operations are over, the land is abandoned as useless or badly damaged. No vegetation is possible in these lands.

(iv) Noise Pollution

Mining and other related operations create heavy noise pollution in and around the quarries.

(v) Deforestation

For operationalising a mine, large tract of forest cover is destroyed. This affects the biodiversity in that region apart from accelerating soil erosion.

The following are the important laws legislated to regulate mining operations in our country.

- i) Wildlife (Protection) Act 1972 (Amended in 1991 and 2002)

- ii) Water (Prevention and Control of Pollution) Act, 1974 (Amended in 1988)
- iii) Forest (Conservation) Act, 1980 (Amended in 1988)
- iv) Environment (Protection) Act, 1986

National Mineral Policy of India, enunciated in 1983 forbids mining operations in the reserve forests for wild life as well as the wild life sanctuaries.

2:05:4 Ways and Means of Preventing Over exploitation of Mineral Resources

The following are the laws legislated to regulate mining operation in our country in order to prevent over - exploitation of minerals and unlawful mining.

- i) Wildlife (Protection) Act 1972 (Amended in 1991)
- ii) Water (Prevention and Control of Pollution) Act 1974 (Amended in 1988)
- iii) Forest (Conservation) Act, 1980 (Amended in 1988)
- iv) Environment (Protection) Act, 1986
- v) National Mineral Policy of India, 1993 (Amended in 2008)

National Mineral Policy, 1993 forbids mining operations in the reserve forests for wildlife as we

as the wildlife sanctuaries. Apart from reaffirming the above, the amended National Mineral Policy, 2008 insist on the following :

- i) Scientific and detailed prospecting is to be carried out in search of mineral resources in sea beds with the help of Remote Sensing Satellite.
- ii) Application of advanced technologies to be used in mining out minerals and modernization of the mining industry to ensure optimal mining plans, and zero waste mining.
- iii) Ensuring that livelihood of tribal people is not adversely affected; appropriate compensation for displaced and affected persons to ensure sustainable development.
- iv) Preservation of wildlife forests and sanctuaries.
- v) Granting mineral prospecting licenses and mining leases to be made transparent.
- vi) Mining leases are to be granted in such a way, not to disturb the eco-system of a region severely.
- vii) Delegating authority to State for mining minerals except coal and atomic minerals and strictly enforcing the same through constant monitoring.

- viii) Permitting export of minerals to other countries only in value added forms.

2:06 Food Resources

Food is the second priority of man in his list of basic needs, the first being water. Food is sustenance. Primitive man fulfilled his food needs by hunting animals. Then he started eating whatever fruits, leaves and roots he could obtain from the forests. Among the living organisms, plants alone are capable of preparing their food by themselves. Animals and human beings depend on plants and other animals for food. Once man left the forests to settle near the river side, he started cultivating crops and domestication of animals to meet his food needs. Today in India, rice, wheat and corn form the staple food of man; fish and sea food as well as milk and meat obtained from live-stock supplement the food needs of man.

2:06:1 Important Sources of Food

1. Agriculture

Agriculture products form the major source of food for man. To increase agricultural productivity, improved irrigation systems were developed; better seeds and improved methods of tillage helped to increase the annual harvest. The use of steam, oil,

gas, or hydro-electric power allowed farms to be mechanized. This not only raised productivity per hectare but also brought rapid expansion in the total land area farmed. Though modern agricultural practices brought a boom in production (Green Revolution) initially, after a significant period of time, the farming lands became infertile and saline. When agricultural production decreases, demand for food escalates, resulting in famine.

The following are three major causes for decrease in food production.

(i) Soil erosion (ii) Desertification (iii) Farmland Conversion.

2. Fisheries and Sea Food

Fish and marine organisms too serve as human food. They belong to the animal food category. Though vegetarian food is gaining popularity at present at global level, yet majority of people in the world like eating fish and sea food as they are rich in protein. Fish and marine resources too are decreasing day by day due to environmental changes, water pollution and over-fishing. This could be avoided by establishing and maintaining fish farms and increase fish production.

3. Live-Stocks

Domestic animals reared in our houses like goat, cow, hen, cock etc. provide meat to us. Apart from these, growing mushroom is also now gaining significant attention. As it contains high level of protein, it is recommended by many.

2:06:2 Food Crisis

In the world, developed countries, rich nations and countries having vast areas of land are not much affected by food crisis. But developing countries like India, Pakistan and Bangladesh as well as most of the under-developed African countries are at present encountering severe food crisis. Most of the nations like Japan, Norway etc. which do not produce food in large scale, manage to meet their food needs by importing from other nations. Though rich nations could afford to import food liberally, under-developed countries do not have the economic power to buy food from other countries to meet the local needs. This has created the problem of uneven distribution of food across the globe.

One of the biggest challenges in independent India is hunger. According to Economic historian **Tirthankar Roy**, the food grain shortage is a gift of

the colonial period to independent India. From 1891 to 1946, the average growth of crop outlay was only 0.4% which translated into an increasing number of poor populations along with overall population. Consequently, per capita availability of food fell and the crisis was severe in over populated Bengal and resulted in the famine of 1943.

In independent India, though food production increases year after year, thanks to five year plans, due to uncontrolled population growth, food demand also goes on increasing making the situation quite difficult to meet both ends. Further we lack adequate infrastructure for storing the food grains as well as good transport facilities to ensure proper distribution of food in all regions of the country. Though India has achieved self-sufficiency in the food front through “Green and White revolutions”, yet we face food crisis due to growing population, inadequate storing warehouses and transportation facilities in all regions, resulting in starvation deaths in some places.

According to Food and Agricultural Organization (FAO) Report of 2021, the onset of the pandemic has made the efforts of making available food to all

citizens even more challenging. The Executive Director of the UN Food Programme, **David Beasley** in 2020 warned global leaders against the looming crisis of hunger. The 2021 Global Hunger Index (GHI) report is a reflection of the same warning. The GHI report came at a time when the world is recovering from the devastation of COVID -19. India slid down to rank 101 out of 116 countries.

In short, we can list the following as the important dimensions of food crisis (i) **Slow Economic growth of the country** (ii) **Inadequate food production** (iii) **Lack of quality warehouses to store food materials and good transport facilities** (iv) **Malnutrition** and (v) **Lack of awareness about ‘balanced diet’**.

2:06:2:01 Ways of solving the Food Crisis

Solving of food crisis has the following two dimensions:

1. Increasing food production
2. Improvement of agricultural management

A. Increasing Food Production:

Food production could be enhanced by

- (a) Promotion of intensive agriculture

- (b) Providing more irrigation facilities
- (c) Updating the barren lands to become fit for farming
- (d) Applying in agriculture latest techniques/ technologies, use of latest implements and equipment to save time and labour, use of high yielding varieties of seeds etc.
- (e) Judicious use of organic and chemical fertilizers; inter-planting of legumes or new nitrogen fixing crops
- (f) Resorting to biological control of pests instead of using destructive chemicals
- (g) Use of eco-friendly insecticides like "Neem Gold" etc.
- (h) More use of soil conservation practices

B. Improvement in Agricultural Management

- (i) Training farmers in the modern techniques and technologies in agriculture
- (ii) Making effort to minimise the cost of production of agricultural produce.
- (iii) Providing easy and low interest loan and insurance facilities to farmers by the government.

- (iv) Planning of crop cultivation by rotation.

2:07 Energy Resources

Today, in all activities of human life, use of energy has become indispensable. Energy is required to fulfill our daily basic needs and also for enjoying the comforts of life. The economic development of a country is judged in terms of the total amount of energy generated and made use of.

Early man discovered fire and its uses. Later on, other sources of energy like timber, animals, agricultural wastes, fossil fuels like coal and petroleum, natural gas etc. were identified and put into use. Today electric power occupies a prime place in human life though other sources of energy are also tapped extensively using modern technologies.

2:07:1 Growing Energy Needs

Demand for energy continues to grow due to many factors such as increasing population, smaller households, expanding industries and occupations, rising household disposable incomes driving more consumption of power, global warming and the consequent long spell of severe summer experienced year after year forcing the continued use of air

conditioners, a variety of new electronic and electrical goods getting introduced in the market, now and then etc. There is a growing demand for energy from all sources - brown coal, gas and petroleum products, and from all sections of the society. In India, except Gujarath almost all states have deficit of electric power to the tune of 20%. Though Central and State Governments have taken tremendous efforts to install different kinds of power stations (Thermal, Hydro and Nuclear), yet they take a gestation period of two to three years before being fully operational. In the intervening period as population increases, the gap between the supply and demand almost remains the same or on the increase.

Tamilnadu Government at present gets electric power to the tune of about 13,000 M.W. daily on an average from different sources like wind mills, its own thermal and hydro electric power stations, power drawn from the Central Pool as its share and power procured from other States for a cost. But the demand hovers around 14,000 M.W. and the peak hour demand touches 15,000 M.W.

State Governments try to manage the power front by adopting various strategies like imposing power cuts, announcing periods of power interruption and

power regulation measures. As a result of this our industries get crippled and production suffers; industrialists and entrepreneurs start migrating to other States; agricultural production falls drastically. This affects the economic growth of the State which leads to shunted economic growth of the country as a whole. To tide over the power crisis, Tamilnadu Government has announced its "Solar Energy Policy - 2012" and initiated measures to achieve a target of generating 3000 MW of solar power per day by 2015. By increasing the number of wind power stations with the support of the private sector, it is aimed to generate 4000 M.V. of power per day. Efforts are also intensified now to develop the power conducting paths from the Northern Grid to the Southern Grid. After a long tussle, with warring groups of people opposing the operation of **Nuclear Power Station at Kudangulam**, it started generating power in its first plant from October 2013 and second plant started production from October 2016, both generating 2000 MV of Power.

The following table illustrates our growth in energy production and continued increase in demand for power consumption.

India's Power Production and Consumption (In Giga Watt-hour)

Year	Net Electricity Generated [Excluding Consumption in Power Station Plus distribution(loss)]	Electricity Sold to Consumers
1970-71	43,724	43,724
1975-76	60,246	60,246
1980-81	82,367	82,367
1985-86	123,106	123,099
1990-91	190,420	190,357
1995-96	277,078	277,029
2000-2001	316,795	316,600
2005-06	412,096	411,887
2006-07	455,964	455,749
2007-08	502,267	501,977
2008-09	527,564	553,995
2009-10	569,723	612,645
2010-11	617,097	694,392
2011-12(P)*	663,392	772,603
Growth Rate (%) From 2010-11 to 2011-12	7.5%	11.26
Total Growth Rate (%) From 1970-71 to 2011-12	6.69%	7.08

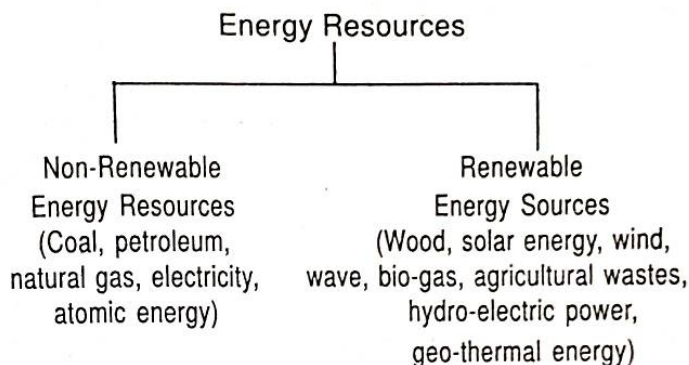
Note :

- 1 Giga Watt-hour = 10^6 K.W.H (or Million Units)
- * Tentative estimates
- From 2008 onwards there exists power

shortage, which is at present about 18%. Energy demand is increasing annually at the rate of 12%

2:07:2 Kinds of Energy Resources

There are different types of energy resources. They may be divided into two major types — (i) Non-Renewable energy resources and (ii) Renewable energy resources.



Energy resources which get depleted on continued large scale use finally become completely exhausted in few decades is called "non renewable energy sources". Eg. fossil fuels like coal, petroleum, natural gas and atomic energy.

Energy resources which are found in unlimited measures in nature and available without any interruption is known as "renewable energy resources". It is also called, "non-exhaustible energy resources", Eg. energy received from the sun, water,

waves, air etc. Of these, we generally used hydro-electric power so far. But water is not available in plenty at all times. Now it is high time to try other alternate sources of renewable energy which are known as '**Non-conventional Sources of Energy**'.

Now, we shall discuss the different forms of these two kinds of energy sources.

2:07:2:01 Non-renewable Energy Sources

1. Fossil Fuels

The remains of plants, trees and animals which decayed and got buried deep in the earth for thousands of years slowly get transformed into fossil fuels. They are found in nature and belong to non-renewable category of fuels. The following are the different kinds of fossil fuels.

i) Crude Oil or Petroleum

Petroleum deposits are found in the river basins and coasts, at a very great depth.

Geo-analysts collect data regarding the nature of soil at different places in the country and identify the areas rich in petroleum. In these places deep bore wells are established to extract petroleum by pumping it from the deep. The petroleum so extracted is transported through lorries and ships to petroleum refineries.

ii) Natural Gas

Natural gas is generally spotted in those places where petroleum wells are located. On burning natural gas, as we get more thermal energy than coal and diesel and also 50 to 70% of less emission of CO_2 . Natural gas is preferred over coal in the production of electricity. The cost of extracting natural gas is also relatively less.

iii) Coal

In the world, Coal is the abundantly available fossil fuel, but the production and consumption of coal greatly affects the environment.

2. Atomic Power

Nuclear Power stations convert the enormous quantity of heat released by splitting an atom of Uranium - 235 or Plutonium - nuclear fission - to generate electricity. The heat released by nuclear fission is transferred to water which in turn is used to produce steam from the water contained in a closed container. The steam thus generated is used to operate the turbines to produce electricity.

The pollution caused by nuclear power stations is relatively less compared to the use of petroleum and coal; the cost of transporting nuclear fuel is also less as compared to coal.

But disposing the nuclear wastes is really a challenging task; thermal pollution is also high. Chances are more for the employees of nuclear power plant as well as those living near it to get affected in health due to constant radiations. In case of accidents, damage will be very severe for men and materials situated within a radius of 100 miles. That is why social acceptance for nuclear power plants is less as compared to other forms of power generations (the struggle of Kudangulam activists can be cited here). Further the cost of establishing and maintaining a nuclear plant is high; availability of fissionable uranium is also less.

2:07:2:02 Renewable Energy Sources

Hydro-electric power, Wind energy, Biomass and Biogas energy, Solar energy, Tidal and wave energy and Geothermal energy are the important renewable energy sources. Among these, the **Hydro-electric power is a conventional renewable energy source; others are known as non-conventional or Alternative source of energy.**

Let us discuss briefly both types of renewable sources of energy

1. Hydro-electric Power

The water deposited in dams built across rivers

is ejected through specially designed pipes to flow out with great force which drives turbines to produce electricity.

Hydro-electric power meets only 20% of the energy needs of the world. The big advantages of hydro-electric power plants are, the cost of production is low; practically no pollution takes place and the technology used is of high quality. Apart from being a renewable source of energy, it helps to control flood, facilitate irrigation and transport. Hydro-electric power resources of India is found to be 4×10^{11} KWH (1KWH = 1 unit of electricity).

Hydro-electric power could be generated only when there is water flow in the river; further, construction cost of dam is very high; destruction of forest, displacement of the locals and loss of biodiversity causes heavy damage to the environment; chances for earthquakes is also high due to depositing of water in the dam.

2. Wind Energy

Among the **non-conventional energy resources**, wind is an important one. Like Sunlight, wind blows through out the world and is also a renewable energy. Wind mills are installed in high wind zones and electricity is produced with the help of strong wind

flow. The specially designed blades mounted on top of the wind mills rotate fast when strong wind blows. This rotation is coupled to a turbine and electricity is produced. Wind mills can also be used to pump water from wells and run flour mills.

Generally wind mills are located in coastal regions, high mountain ranges or wide open valleys where wind flow is stronger, not less than 15 K.M. per hour. Tamilnadu is leading in producing electricity by wind energy followed by Gujarat. In Tamilnadu with the help of **Danish International Agency**, wind farms are established at **Kayathar** (Thoothukudi Dt.), Muppandal and Mullaikadu (Kanyakumari Dt.). The Ministry of Non-conventional energy encourages the development of wind energy.

Merits

1. Wind energy is a renewable resource.
2. Wind required for running the wind mills is free of cost.
3. There is no environmental pollution.
4. Though initial construction cost is high, maintenance cost is very less.
5. Cost of production is very low.
6. In wind mill areas rural industries thrive, offering employment opportunities to rural people.

Demerits

- i) Windmills can not operate in low wind zones.
- ii) Wind flow will not be strong, through out the year and hence power generation is seasonal.
- iii) Windmills get damaged during storms
- iv) Though large amount of electricity is generated during storms, it can not be stored.
- v) Windmills may spoil the beauty of the natural scenery of a place.

3. Biomass and Biogas Energy

Fermentation of starch in plants like corn and tapioca produces alcohol that can be used to run automobiles. From particular type of plants, petroleum like product is obtained. Further traditionally in our houses using sawdust, straw, paddy husk, rice bran, cow dung and domestic wastes thermal energy was produced.

When organic wastes of plants and animals decays, biogas (Methane) is released which could be used as an alternative fuel. After producing methane, the residual wastes can be used as manure for plants. Burning of biogas causes no environmental pollution. In rural areas cow dung is used to produce gobar gas and this gas is utilized for cooking and lighting.

Merits

- i) As energy is produced from organic wastes, which are available in abundance in rural areas, production cost is cheap.
- ii) As most of the waste are used for energy production and the rest as manure, there is practically no environmental pollution.

Demerits

- i) This method is not viable in all areas except some rural pockets.
- ii) There are practical difficulties in collecting, storing and using for energy production.
- iii) Rural people do not show much interest in using biomass energy.

Biomass energy meets 20% of the world energy requirements. This non-conventional source of energy proves to be a helping hand for developing countries to tide over energy deficits. This energy could be used in many ways.

4. Solar Energy

Sun is the primary source of energy which could be tapped from solar radiation. During the olden days, solar energy was primarily used for drying clothes, and grains, extracting salt by evaporating sea water and preserving food. At present we are

using solar energy for various purposes due to scientific advancement. In general, solar energy can be used in two methods : (i) Solar photo voltaic power system (ii) Solar thermal energy system.

(i) Solar Photo Voltaic Power System

In this method, solar energy is transformed into electricity through solar cells made up of silicon. By fixing silicon cells on roof top of building terraces, we can generate electricity, that could be stored in silicon batteries and used to operate machines and instruments in industries. The obtained solar energy could also be used (with the help of invertors) to operate house lights, radio, television, pump set and other electrical appliances.

Remote villages which can not get electricity due to transmission difficulties can produce electricity for houses and street lights in this method. This system is used to supply electricity for the electric fences to protect the crops from man and animals, operating railway signals, light houses and offshore platforms. Electricity required for satellites in space is produced from the solar panels attached to them.

(ii) Solar Thermal Energy System

By using Solar Thermal Collectors, the heat received from solar radiation could be transformed

into thermal energy. By concentrating solar radiation, solar thermal devices like solar cookers, water heaters, desalination systems, solar refrigerators, solar dryers etc. can be operated.

Advantages of Solar Energy

- Solar energy is a renewable non-conventional resource.
- Solar energy is available throughout the year in equatorial and tropical regions.
- There is no environmental pollution.
- Input for solar devices costs nothing.
- Cost of production is, though high initially, over the years due to cost-free maintenance, it proves cheaper; further energy can be stored in batteries for after use.

Demerits

- To develop solar silicon panels, large amount of capital is required.
- During cloudy and rainy days production of electricity and heating are affected.
- Cadmium used in the production of solar cells is poisonous.

5. Tidal and Wave Energy

According to the position of the sun and moon, due to the change in gravitational forces there is a

rise and fall in sea level. Power can be produced if the difference between the high and low tide level is more. Sea water is allowed to flow with high force into a reservoir on the seashore during high tide. The water flow is used to rotate the turbines and electricity is produced. The stored water is allowed to return into the sea again through the same turbine during the low tide and here again electricity is generated. This method is used in America at the **Bay of Canada**, in **Cambay** and **Kutch** (in India) to produce electricity with the help of tides.

Sea waves are in constant motion which is used for producing energy. By installing floating turbines against waves we can produce electricity. In shallow seas, turbines fixed on platforms are rotated by wave motion and electricity is produced.

Merits

- Electricity produced by using waves and tides causes no environmental pollution.
- It is a renewable and non-conventional energy source and production of electricity is cheap.
- It can be implemented in regions having big seas.

Demerits

- The amount of electricity produced by this method is very low.

- ii) This method can not be implemented in regions which do not have large waves and tides.
- iii) During storms, electricity producing equipments may get damaged.

6. Geothermal Energy

In the interior of earth, rocks are in the molten form with high temperature and pressure. In regions where volcanic activity is high, hot springs are found. The steam released from these hot springs could be used to rotate the turbines to produce electricity. [Note : Here it could be recalled that in the hot spring near Yamunothri of Himalayas, potatoes are boiled and given as prasadam]. In Java Islands of Indonesia, and in some other regions, water is artificially pumped into the earth and this water is converted into steam due to the high temperature in the interior of the earth. This steam is used for generating electricity.

Merits

- i) This is a renewable non-conventional source of energy.
- ii) Geothermal energy is ecofriendly and pollution free.
- iii) Electricity produced in this method is very cheap.

Demerits

- i) The construction and maintenance cost of the plant is high in volcanic regions.
- ii) Equipments and structures will be lost during earth quakes and volcanic eruptions.
- iii) At times, harmful sulphur gas may come out with, from the interior of the earth steam.

Conclusion

In this unit, we have discussed various kinds of natural resources viz. land, forest, water, minerals, food and energy and the problems associated with them and adverse effects of over exploitation of natural resources.

Review Questions :

1. Mention the different kinds of natural resources. (C) [Ans: 2:01]
2. Discuss the factors that affect land resources. (A) [Ans. 2:02:1 + 2:02:1:01 to the end of 2:02:1:03]
3. What is soil erosion? State the reasons for it. (C) [Ans. In 2:02:1 under 1 and 2]
4. Mention the effects of soil erosion (B) [Ans. In 2:02:1:01, those under (C)]
5. What is the meaning of 'soil fertility'? What are the reasons for its degradation? (B) [Ans. 2:02:3 + 2:02:3:01]